

National Institute of Technology Kurukshetra
Fuzzy Systems and Evolutionary Computing (AIPC206) (AIML Branch)
Mid-Term Examination-I, Even Semester, 2025-26

Time: 50 Minutes

Total Marks: 15

Note: Attempt all the questions. If require any missing data; then choose suitably.

Q1. a) Draw the following member functions with x axis as universe of discourse and y axis as membership value. i) Triangle ($x: 10, 40, 100$) ii) Max (Min ($\frac{x-5}{30-5}, 0.8, \frac{80-x}{80-60}, 0$), 0) iii) Gaussian ($x, 50, 20$) iv) Sigmoid ($x, 1, -5$)

[2 marks]

b) A fuzzy set is defined as $\underline{1}$ (Fuzzy 1) = $\{0.5/0+1.0/1+0.5/2\}$. Another fuzzy set $\underline{2}$ (Fuzzy 2) is defined as $\{0.2/0+0.5/1+1/2+0.6/3+0.1/4\}$. Add $\underline{1}$ and $\underline{2}$ using extension principle.

[2 marks]

c) Prove or disprove if operator $N(x) = (1 - x^4)^{\frac{1}{4}}$ can be used as a negation operator and whether this operator follows Involution principle.

[2 marks]

Q2. a) A high speed rail monitoring device is used to find and detect the speed of a rail from the angle of deflection. The high speed is a fuzzy set B defined as $\{0.2/1 + 0.4/2 + 0.7/3 + 1.0/4 + 0.8/5\}$, where 1, 2, 3, 4, 5 are normalized units for speed. Another fuzzy set high deflection (A) is defined as $\{0.3/10 + 0.8/20 + 1.0/30 + 0.7/40\}$, where 10, 20, 30, 40 are in micro radians. The rule is: "high angle of deflection is high speed". Given the antecedent "nearly high deflection" (A') = $\{0.3/10 + 1.0/20 + 0.9/30 + 0.6/40\}$, find the corresponding (B').
Use material implication.

[3 marks]

b) Given, Premise 1 (rule): if x is A and y is B then z is c , Premise 2 (fact): x is A' and y is B' . i) Prove consequence " z is c " = $[A' \circ (A \rightarrow C)] \wedge [B' \circ (B \rightarrow C)]$. ii) Also prove that $\mu_c(z) = w_1 \wedge w_2 \wedge \mu_c(z)$. Where w_1 and w_2 are degree of compatibility between A, A' and B, B' respectively.

[3 marks]

Q3. Discuss the properties of T-norm operators. Explain any four T-norm operators with example.

[1+2=3 marks]